22.5.2 Coal and Petroleum

Coal and Petroleum are **natural resources**. They are called **fossil fuels** as they are formed from the degradation of biomass buried deep under the earth millions of years ago.

DO YOU KNOW?

India is the third largest consumer of crude oil in the world, after the United States and China.

Coal is used for **generation of electricity** at Thermal power plants. **Petroleum** also known as **crude oil** is processed in oil refineries to produce **petrol** and **diesel** which are used to run automobiles, trucks, trains, ships and airplanes etc. **Kerosene** and **LPG** (Liquefied Petroleum Gas) obtained from petroleum is used as domestic fuel for cooking food.

The coal and petroleum reserves can get exhausted if we continue using them at a rapid rate. The formation of these fossil fuels is a very slow process and takes very long period of time for renewal.

22.5.3 Steps to Conserve Coal and Petroleum Resources

It is necessary to conserve or save coal and petroleum resources for the future use, which can be done by reducing their consumption.

(i) If electricity is saved, it will inturn reduce the use of coal
(ii) Using bicycle for covering short distances instead of using cars, scooters or motorcycles
(iii) Using pressure cooker can reduce the consumption of kerosene and LPG while cooking food. Solar cooker and solar heaters can be used wherever possible
(iv) Motor vehicles should be designed with fuel efficient engines to increase efficiency and also reduce air pollution

Case study of Taj Mahal

The Taj Mahal is one of the seven wonders of the world and is located in Agra, Uttarpradesh. It is built with white marble. The Mathura oil refinery owned by Indian Oil Corporation present around this area produce sulphur and nitrogen oxides. The white marble became yellow due to air pollution. The Government of India has set up emission standards around the monument to protect it from the damage.

22.6 Non-Conventional (Alternative) Energy Resources

The energy crisis has shown that for sustainable development in energy sector we must conserve the non-renewable conventional resources from its rapid depletion and replace them by non-polluting, renewable sources which are environmentally clean.

Efforts are made to develop new sources of energy which is called non-conventional sources of energy. It would provide greater initiative to local people who could assess their needs and resources and plan a strategy that could be useful to them.

22.6.1 Solar Energy

Solar energy is the **energy obtained from the sun**. The sun gives out vast amount of light and heat. It is only a little less than half (47%) of solar energy which falls on the atmosphere reaches the earth's surface. If we could use just a small part of this energy it would fulfill all the country's need for power. Solar energy has advantages and also certain limitations.

Solar Energy Devices

The energy from the sun can be harnessed to provide power. The various devices used for harnessing sun's energy are called solar energy devices.

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